

**SIMATS ENGINEERING**  
**ASSESSMENT TOOL 2 : Scenario-Based**

|                     |                                        |                     |         |
|---------------------|----------------------------------------|---------------------|---------|
| <b>COURSE NAME:</b> | Cloud Computing and Big Data Analytics | <b>COURSE CODE:</b> | CSA1501 |
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**COURSE OUTCOME COVERED**

**CO2:** *Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)*

Assessment Tool Weightage: 20%

Dave's Taxonomy: Manipulation (Level 2)  
Question Count: 5

**Total Marks: 25**

**Code of Conduct**

I P.Poorna Chandra(192472287) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: P.Poorna Chandra

**Assessment Tasks**

**Scenario-Based Questions**

- 1.A media company wants to migrate its video streaming platform to the cloud. The system must handle millions of concurrent users with minimal buffering. Design a cloud architecture using scalability and caching techniques. Explain how load balancing and auto-scaling can ensure smooth streaming. Discuss cost optimization strategies for such high-demand workloads.
- 2.A fintech startup is deploying applications using containers across multiple cloud providers. They face challenges in maintaining consistency and managing deployments. Propose a solution using container orchestration and multi-cloud strategies. Explain how service discovery and scaling can be handled efficiently. Highlight the benefits and risks of a multi-cloud deployment model.
- 3.An e-learning platform experiences data loss due to accidental deletion in the cloud. The organization needs a robust backup and disaster recovery strategy.

Design a solution using cloud-native backup, replication, and versioning.  
 Explain how recovery time objective (RTO) and recovery point objective (RPO) are achieved.  
 Provide steps to ensure business continuity during failures.

4.A government agency is adopting a hybrid cloud model for sensitive and non-sensitive data.  
 Some applications must remain on-premises due to compliance requirements.  
 Design a hybrid architecture that ensures secure communication between environments.  
 Explain how identity management and encryption should be implemented.  
 Discuss challenges in managing hybrid cloud environments.

4. A SaaS provider needs to ensure high availability and fault tolerance for its application.  
 The system must continue operating even if one region goes down.  
 Propose a multi-region deployment strategy in the cloud.  
 Explain how data replication and failover mechanisms should be configured.  
 Discuss monitoring and alerting strategies for proactive issue detection.

### Scenario-Based Rubric

| Criteria                         | Weightage | Level 4 - Exemplary                                         | Level 3 - Proficient                 | Level 2 - Developing          | Level 1 - Beginning              |
|----------------------------------|-----------|-------------------------------------------------------------|--------------------------------------|-------------------------------|----------------------------------|
| Understanding of Scenario        | 20%       | Demonstrates complete understanding of the scenario context | Good understanding with minor gaps   | Partial understanding         | Fails to understand the scenario |
| Identification of Risks / Issues | 20%       | Identifies all major issues correctly                       | Identifies most issues               | Limited issue identification  | Misses major issues              |
| Application of Concepts          | 25%       | Applies virtualization concepts effectively                 | Applies concepts with minor mistakes | Partially correct application | Weak application                 |
| Decision Justification           | 20%       | Decisions are well justified and technically strong         | Reasonable justification             | Weak justification            | No useful justification          |
| Clarity                          | 15%       | Clear and well-organized response                           | Generally clear                      | Somewhat unclear              | Poorly presented                 |

## 1. A Media Company Wants to Migrate Its Video Streaming Platform to the Cloud

### Answer

#### Scenario Analysis

A media company provides video streaming services to millions of users. During peak hours, a large number of users watch videos simultaneously. The cloud architecture must ensure high performance, low buffering, and continuous availability.

#### Cloud Architecture Design

The architecture includes:

- **Users** access the platform through the Internet.
- A **Load Balancer** distributes user requests across multiple application servers.
- **Auto-Scaling Groups** automatically add or remove servers depending on traffic.
- A **Content Delivery Network (CDN)** caches videos at edge locations closer to users.
- **Cloud Storage** stores video files.
- **Database Replication** maintains user profiles and streaming history.
- **Monitoring Services** continuously monitor CPU, memory, and network usage.

#### Architecture Flow:

Users

|

Load Balancer

|

Auto Scaling Group

|

Application Servers

|

Cloud Storage + Database

|

CDN (Cached Videos)

#### Load Balancing

A load balancer distributes incoming requests evenly among multiple servers.

#### Benefits

- Prevents server overload.

- Improves response time.
- Increases application availability.
- Supports millions of concurrent users.

### **Auto Scaling**

Auto Scaling automatically increases or decreases virtual machines based on demand.

Example:

- During live sports events, additional servers are launched.
- After traffic decreases, unused servers are terminated.

### **Benefits**

- Maintains smooth streaming.
- Reduces buffering.
- Optimizes resource usage.

### **Caching Techniques**

Caching stores frequently requested videos closer to users.

Methods include:

- CDN edge caching
- Browser caching
- Application caching

### **Benefits**

- Faster video delivery
- Lower latency
- Reduced bandwidth usage
- Less load on origin servers

### **Cost Optimization Strategies**

- Use Auto Scaling to avoid paying for idle servers.
- Store old videos in low-cost cloud storage.
- Use Reserved or Spot Instances where appropriate.
- Compress videos efficiently.
- Cache popular content using a CDN.

### **Conclusion**

A combination of **Load Balancing, Auto Scaling, CDN caching, and cloud storage** provides high scalability, low buffering, and cost-efficient video streaming for millions of users.

## 2. A Fintech Startup is Deploying Applications Using Containers Across Multiple Cloud Providers

### Answer

#### Scenario Analysis

A fintech startup deploys applications across AWS, Azure, and Google Cloud. They require consistent deployments, high availability, and simplified application management.

#### Proposed Solution

Deploy containers using **Docker** and manage them using **Kubernetes**.

Architecture:

Developers

|

Docker Containers

|

Kubernetes Cluster

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AWS | Azure | Google Cloud

#### Container Orchestration

Kubernetes automates:

- Container deployment
- Scaling
- Load balancing
- Self-healing
- Rolling updates

If one container fails, Kubernetes automatically creates a new one.

#### Service Discovery

Each service receives a unique DNS name.

Example:

payment-service

login-service

notification-service

Containers communicate without knowing IP addresses.

### **Scaling**

Horizontal Pod Autoscaler increases or decreases container replicas according to CPU and memory utilization.

Example:

Traffic increases:

2 Pods → 10 Pods

Traffic decreases:

10 Pods → 2 Pods

### **Benefits of Multi-Cloud**

- High availability
- Vendor independence
- Disaster recovery
- Better global performance
- Flexible pricing

### **Risks**

- Complex management
- Different cloud APIs
- Higher networking costs
- Security policy differences
- Data synchronization challenges

### **Conclusion**

Using **Docker, Kubernetes, service discovery, and auto scaling** provides a consistent, reliable, and scalable multi-cloud deployment.

## **3. An E-learning Platform Experiences Data Loss Due to Accidental Deletion**

### **Answer**

#### **Scenario Analysis**

Student records, course materials, and examination data have been accidentally deleted. The organization requires rapid recovery with minimal downtime.

## **Backup and Disaster Recovery Design**

The solution includes:

- Automatic cloud backup
- Cross-region replication
- Object versioning
- Database snapshots
- Disaster Recovery site

Architecture:

Application

|

Cloud Storage

|

Automatic Backup

|

Cross Region Replication

|

Disaster Recovery Site

## **Cloud Native Backup**

Daily automatic backups are stored securely.

Benefits:

- Easy recovery
- Long-term storage
- Minimal manual work

## **Replication**

Copies of data are stored in another cloud region.

Benefits:

- Protection against regional failures
- High availability

## **Versioning**

Every modification creates a new version of a file.

If a file is deleted accidentally, previous versions can be restored immediately.

### **Recovery Time Objective (RTO)**

RTO is the maximum acceptable downtime.

Example:

RTO = 15 minutes

Recovery begins immediately after failure.

### **Recovery Point Objective (RPO)**

RPO indicates the maximum acceptable data loss.

Example:

Backups every 5 minutes

RPO = 5 minutes

### **Business Continuity Steps**

- Automatic backups
- Regular disaster recovery testing
- Multi-region replication
- Failover procedures
- Continuous monitoring

### **Conclusion**

Backup, replication, and versioning ensure fast recovery, minimal data loss, and uninterrupted learning services.

## **4. A Government Agency is Adopting a Hybrid Cloud Model**

### **Answer**

### **Scenario Analysis**

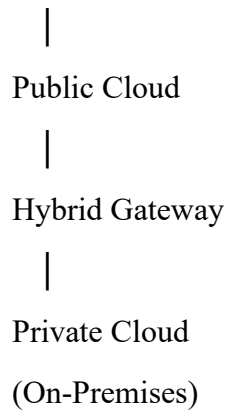
Sensitive citizen information must remain on-premises, while public applications can be deployed in the cloud.

### **Hybrid Cloud Architecture**

Users

|

Secure VPN



Sensitive databases remain inside the government data center.  
Public services such as websites run in the public cloud.

### **Secure Communication**

Secure communication uses:

- VPN
- TLS/SSL encryption
- Private network connections
- Firewalls

This protects data while moving between environments.

### **Identity Management**

Implement:

- Single Sign-On (SSO)
- Multi-Factor Authentication (MFA)
- Role-Based Access Control (RBAC)
- Identity Federation

These ensure only authorized users access resources.

### **Encryption**

Data should be encrypted:

#### **At Rest**

- AES-256 encryption

#### **In Transit**

- TLS/SSL encryption

### **Challenges**

- Complex management
- Higher operational cost
- Data synchronization
- Security policy consistency
- Network latency

## **Conclusion**

A hybrid cloud combines the security of private infrastructure with the scalability of the public cloud while meeting government compliance requirements.

## **5. A SaaS Provider Needs High Availability and Fault Tolerance**

### **Answer**

### **Scenario Analysis**

A SaaS application must continue operating even if an entire cloud region fails.

### **Multi-Region Deployment Strategy**

Deploy the application across multiple cloud regions.

Example:

Region A (Primary)

↓

Region B (Secondary)

↓

Region C (Backup)

Traffic is automatically redirected if one region becomes unavailable.

### **Data Replication**

Data should be replicated continuously between regions.

Methods include:

- Synchronous replication (critical databases)
- Asynchronous replication (less critical workloads)

Benefits:

- Prevents data loss
- Supports disaster recovery

### **Failover Mechanism**

When Region A fails:

- Health checks detect failure.
- DNS or Load Balancer redirects traffic.
- Region B becomes active automatically.

Users experience minimal downtime.

### **Monitoring and Alerting**

Use cloud monitoring tools to monitor:

- CPU utilization
- Memory usage
- Disk space
- Network latency
- Application health

Alerts should be sent through:

- Email
- SMS
- Mobile notifications

## **Benefits**

- High availability
- Fault tolerance
- Disaster recovery
- Continuous service
- Better customer satisfaction

## **Conclusion**

A **multi-region architecture** with **continuous data replication**, **automatic failover**, and **proactive monitoring** ensures that the SaaS application remains available even during regional outages, delivering reliable and uninterrupted service.